



Spatiotemporal Dynamics and Multi-Factor Drivers of Land Surface Temperature in Andhra Pradesh, India (2020-2024): Integrating MODIS-Derived Indices, Urban Heat Islands, Hotspot Analysis and Trend Detection

**M.R Goutham ^{a*}, Suneel Kumar Duvvuri ^b
and Umamahesh Gouda ^a**

^a Department of Geology, Government College (Autonomous) Rajahmundry, AP, India.

^b Department of Computer Science, Government College (Autonomous) Rajahmundry, AP, India.

Authors' contributions

This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

Land Surface Temperature (LST) is recognized as a primary indicator of regional thermal shifts precipitated by rapid urban expansion and land-use modifications. In this research, a comprehensive five-year longitudinal assessment (2020-2024) of LST dynamics across Andhra

*Corresponding author: E-mail: gouthammr@gcrjy.ac.in

Pradesh, India, was conducted. The methodology involved the integration of MODIS-derived products with topographic and land-cover datasets, processed via the Google Earth Engine platform. The temporal analysis indicated that the annual mean LST underwent a distinct reduction in 2022, falling by 0.82 °C compared to 2021 levels. Following this decline, a recovery phase was observed, culminating in a marginally significant monotonic increasing trend, as evidenced by the Mann-Kendall test results ($S = +10$, $Z = 2.205$, $p = 0.051$). Statistical evaluation through Pearson correlation for the year 2022 identified the Normalized Difference Built-up Index (NDBI) as the predominant positive driver of temperature ($r = +0.794$). Conversely, mitigating influences were attributed to the Normalized Difference Vegetation Index (NDVI, $r = -0.654$) and the Normalized Difference Water Index (NDWI, $r = -0.541$). Topographic factors also played a role in temperature regulation, with slope ($r = -0.504$) and elevation ($r = -0.201$) showing negative correlations with LST. Regarding localised climate phenomena, Urban Heat Island (UHI) intensities were found to be modest yet persistent, ranging between 0.006 °C and 0.010 °C. The application of Getis-Ord G_i^* hotspot analysis further validated the existence of statistically significant thermal clustering within primary urban centers. Throughout the study period, built-up land covers consistently recorded the highest thermal values. It was concluded that urban growth exerts a dominant influence on regional warming, often offsetting the cooling contributions of vegetation and terrain. These findings offer a critical empirical basis for regional heat-mitigation and urban planning within the context of rapidly developing Indian states.

Keywords: Land surface temperature; MODIS; urban heat island; NDBI; Mann-Kendall trend; Getis-Ord G_i^* ; Andhra Pradesh; Google earth engine.

1. Introduction

Global land surfaces have undergone profound transformations due to rapid urbanization, where the replacement of permeable natural substrates with impervious materials has led to increased absorption and re-emission of solar radiation. This process results in higher temperatures within built-up zones compared to rural peripheries, a condition identified as the Urban Heat Island (UHI) effect (Aflaki et al., 2017), (Santamouris & Vasilakopoulou, 2023). Such localized warming is associated with increased cooling energy requirements, diminished air quality, and escalated public health hazards, particularly during periods of extreme heat (Karimi et al., 2021). In tropical contexts like India, where demographic shifts and economic growth accelerate urban sprawl, UHI effects exacerbate existing climate sensitivities, necessitating regional monitoring for evidence-based policy and sustainable development (Pradeep Kumar et al., 2023; Antil et al., 2025).

Located in southeastern India and spanning roughly 163,000 km², Andhra Pradesh serves as a primary example of these environmental shifts. The state features a complex topography, ranging from coastal lowlands to the Eastern Ghats and a tropical monsoon climate. Significant urban expansion has been recorded in hubs such as Visakhapatnam, Vijayawada, and Tirupati, particularly following the state's

bifurcation in 2014 (Bala et al., 2024). The 2020-2024 study period is of particular interest as it covers the era of COVID-19 lockdowns (2020-2021), which saw a temporary decline in anthropogenic activity, followed by a phase of economic resurgence and rapid infrastructure development (Lele et al., 2021), (Maithani et al., 2020).

1.1 Remote Sensing and Biophysical Drivers

Remote sensing technologies, specifically those utilizing MODIS datasets, are established as effective tools for monitoring Land Surface Temperature (LST) due to their spatial coverage and high temporal frequency (Nayak et al., 2023). Research across the Indian subcontinent has repeatedly indicated a positive correlation between LST and built-up indices (e.g., NDBI), while conversely noting negative relationships with vegetation (NDVI) and moisture indices (NDWI) (Mohammad & Goswami, 2023), (Nikkala et al., 2022). For instance, multi-city investigations using MODIS-derived LST have linked significant warming trajectories to urban growth, while studies in semi-arid regions have pinpointed NDBI as a primary driver of elevated surface temperatures (Galodha & Gupta, 2021).

The lockdown measures associated with COVID-19 provided a rare baseline for

isolating human-induced thermal impacts; several reports documented temporary LST declines of 2-6 K in urban and industrial sectors following the reduction of emissions (Maithani et al., 2020). Furthermore, the application of sophisticated spatial statistics, including Getis-Ord G_i^* for hotspot identification and Mann-Kendall tests (Sai et al., 2025) for trend evaluation, has significantly advanced the depth of UHI research (Cafaro et al., 2024). However, exhaustive state-level research that synthesizes biophysical, human, and topographic factors with UHI intensity and non-parametric trend analysis remains sparse for Andhra Pradesh (Islam et al., 2024). This research gap is significant in light of the state's diverse physiography and recent post-pandemic urban rebound.

1.2 Research Objectives

The present study addresses these identified gaps by utilizing Google Earth Engine to evaluate MODIS-derived LST and associated indices from 2020 to 2024. The primary objectives are as follows:

1. To quantify the spatiotemporal LST configurations across Andhra Pradesh during the 2020-2024 timeframe.
2. To analyze the comparative impact of vegetation (NDVI), water bodies (NDWI), urbanization (NDBI), and terrain characteristics (elevation and slope).
3. To delineate UHI intensity and locate statistically significant thermal hotspots.
4. To determine monotonic trajectories using non-parametric Mann-Kendall testing.

The results demonstrate that urbanization serves as the primary catalyst for regional warming, offering empirical insights for the development of heat-resilient urban strategies in expanding Indian states.

2. Literature Review

The research on Land Surface Temperature (LST) and Urban Heat Island (UHI) effects within India has experienced a significant surge since 2015, facilitated by the enhanced availability of satellite imagery from the MODIS and Landsat platforms (Nayak et al., 2023). Initial research primarily targeted major metropolitan areas, establishing definitive connections between the proliferation of impervious surfaces and the emergence of thermal anomalies (Mohammad &

Goswami, 2023). Furthermore, nationwide assessments utilizing long term MODIS records have identified statistically significant nighttime warming across urban centers, a phenomenon largely ascribed to the expansion of built-up areas (Nayak et al., 2023).

In the context of southern India, focused investigations into Hyderabad and its surrounding territories have reinforced the positive correlation between the Normalized Difference Built-up Index (NDBI) and LST, while simultaneously highlighting the cooling potential of urban greenery (Pradeep Kumar et al., 2023). Observations of seasonal fluctuations have further revealed that UHI intensity typically peaks during the summer months, driven by diminished evapotranspiration rates (Galodha & Gupta, 2021).

The COVID-19 pandemic served as an unprecedented natural experiment, leading to a myriad of studies on short-term environmental recovery. Notable LST reductions, reaching several degrees Celsius, were documented in industrial zones and megacities; these shifts were directly linked to the cessation of vehicular traffic and industrial operations (Lele et al., 2021), (Maithani et al., 2020). Such findings emphasized the susceptibility of anthropogenic thermal signatures to rapid mitigation.

Methodological frameworks in this field have evolved to integrate multiple spectral indices, including the Normalized Difference Vegetation Index (NDVI) for biomass assessment, the Normalized Difference Water Index (NDWI) for surface moisture, and NDBI for urban density (Mohammad & Goswami, 2023). Pearson correlation analyses frequently identify NDBI as the most significant positive forcing agent ($r > 0.7$), which stands in direct contrast to the negative associations observed with NDVI (Nikkala et al., 2022). Additionally, spatial clustering techniques, such as the Getis-Ord G_i^* statistic, have been successfully employed to pinpoint enduring thermal hotspots in core urban areas (Galodha & Gupta, 2021)(Baeza-González & Kamakura, 2025)(Peeters et al., 2015), while Mann-Kendall tests applied to localized time series have frequently detected marginally significant warming trajectories (Nayak et al., 2023)(Cui et al., 2021).

Despite these advancements, the influence of topography-specifically elevation-related lapse rates and slope orientation-remains an area of

limited exploration in coastal states, where such factors may significantly moderate LST (Pradeep Kumar et al., 2023). Currently, there is a lack of research that concurrently maps UHI intensity, Gi* hotspots, multi-year land-use trends, and post-pandemic thermal recovery patterns within regions like Andhra Pradesh (Nikkala et al.,

2022). While existing literature confirms that urbanization is the primary driver of LST in India, it also highlights an urgent requirement for integrated, state-level analyses that synthesize diverse environmental drivers with advanced statistical modelling—a gap that the present study seeks to address.

Table 1. Summary of recent (post-2015) state-of-the-art studies on LST and UHI in India

Study	Year	Location	Data/Source	Key drivers/Analysis	Major findings
(Good et al., 2017)	2017	Global	ATSR LST CDR (17+ yrs) & T2m	LST–air temperature comparison (day/night)	Night LST matches Tmin well; validates satellite LST
(Maithani et al., 2020)	2020	Dehradun	LST during lockdown	Anthropogenic reduction	Significant LST drop in urban/industrial zones
(Lele et al., 2021)	2021	Terrestrial ecosystems	LEO-GEO satellites	COVID lockdown impact	Temporary cooling due to reduced activity
(Galodha & Gupta, 2021)	2021	Global (incl. India context)	LCZ & LST	Local climate zones	Distinct intra-city UHI variations
(Song et al., 2021)	2021	China	MODIS LST (2003–2019), climate & vegetation data	Seasonal & interannual trend analysis; driver attribution	Spring–winter warming dominant; air temperature & vegetation control LST
(Nikkala et al., 2022)	2022	Visakhapatnam (AP)	Landsat-8	LST correlation with LULC	Strong NDBI-LST link in coastal urban area
(Nayak et al., 2023)	2023	Major Indian cities	MODIS LST (2000-2023)	Urban expansion, long-term trends	Significant nighttime warming; SUHI intensity increasing
(Pradeep Kumar et al., 2023)	2023	Semi-arid AP regions	LST & NDBI	Urbanization, climate change	NDBI dominant positive driver; thermal comfort zones identified
(Mohammad & Goswami, 2023)	2023	Metropolitan cities	Multiple indicators	SUHI/SCUI quantification	Combined heat/cool island metrics; urbanization primary cause
(N et al., 2025)	2025	India (with global comparison)	Multi-source satellite data	UHI mapping with ML/DL methods	Gaps in data, scale issues, proposed AI driven methodology for monitoring UHI

The chronological progression of the literature is summarized in Table 1, tracking the transition from long-term MODIS-based trend evaluations to recent analyses of pandemic-induced lockdown consequences and sophisticated correlation modeling. However, it is observed that existing research seldom synthesizes topographic variables and spatial hotspot clustering at a state-wide scale for Andhra Pradesh. Consequently, a comprehensive integration of these multifaceted drivers remains a notable lacuna in the current body of work.

3. Material and Methods

3.1 Study Area

The study area (Fig. 1) encompasses Andhra Pradesh, situated on India's southeastern littoral between 12.4°-19.9°N and 76.7°-84.7°E. Spanning approximately 162,975 km², the region is characterized by diverse physiography, transitioning from the Bay of Bengal coastal plains to the Eastern Ghats, where altitudes exceed 1,600 m. A tropical monsoon climate prevails, with average annual temperatures ranging from 28 to 32 °C and precipitation levels between 800 and 1,200 mm, concentrated during the southwest monsoon (June-September). Since the state's bifurcation in 2014, major urban centers, particularly Visakhapatnam, Vijayawada-Guntur, and Tirupati, have undergone significant demographic and infrastructural expansion. For spatial consistency, the state's administrative boundaries were integrated as vector assets within Google Earth Engine (GEE).

3.2 Data Acquisition

All computational tasks were executed within the GEE environment to facilitate reproducibility and high-performance processing. The following primary datasets were utilized for the 2020-2024 observation period:

- **Thermal data:** MODIS/Terra (MOD11A2 v061) 8-day LST at 1 km resolution. The daytime LST band was selected to evaluate peak thermal stress.
- **Vegetation dynamics:** MODIS/Terra (MOD13A2 version 061) 16-day NDVI at 1 km resolution.
- **Surface reflectance:** MODIS/Terra (MOD09A1 version 061) 8-day composites at 500 m resolution, serving as the basis for NDBI and NDWI calculations.

- **Digital elevation model:** USGS SRTMGL1 version 003 Digital Elevation Model (30 m resolution), resampled to 1 km.
- **Land use/Land cover:** ESA World Cover v200 (10 m resolution) for 2021, aggregated to a 1 km grid.

3.3 LST Processing and Temporal Compositing

The MOD11A2 daytime LST values were subjected to a 0.02 scale factor and converted to Celsius. To mitigate cloud interference and ensure data quality, 8-day composites were averaged into annual mean LST composites for each year from 2020 to 2024. Spatial masking was applied using the vector boundary of Andhra Pradesh.

3.4 Derivation of Spectral Indices

The NDVI was retrieved from the MOD13A2 product, scaled by 0.0001, and annual means were computed from 16-day composites. For moisture and urban density assessment, surface reflectance bands from the MOD09A1 product were utilized. The Indices were computed as follows:

$$NDWI = \frac{(Green - SWIR1)}{(Green + SWIR1)} = \frac{(Band\ 4 - Band\ 6)}{(Band\ 4 + Band\ 6)} \quad (1)$$

$$NDBI = \frac{(SWIR1 - NIR)}{(SWIR1 + NIR)} = \frac{(Band\ 6 - Band\ 2)}{(Band\ 6 + Band\ 2)} \quad (2)$$

Annual mean composites were subsequently generated for both metrics.

3.5 Topographic Characterization

Terrain elevation was extracted from the SRTMGL1 DEM using bilinear interpolation to match the 1 km resolution. Slope gradients (in degrees) were derived using the `ee.Terrain.slope()` algorithm within the GEE environment.

3.6 Urban Heat Island (UHI) Quantification

UHI intensity was defined as the thermal deviation between built-up areas and the rural/non-urban baseline. Non-urban pixels were identified by masking the "Built-up" category (Class 50) of the ESA WorldCover dataset. The mean LST of these non-urban zones was subtracted from the total LST distribution to produce annual spatial maps of UHI intensity.

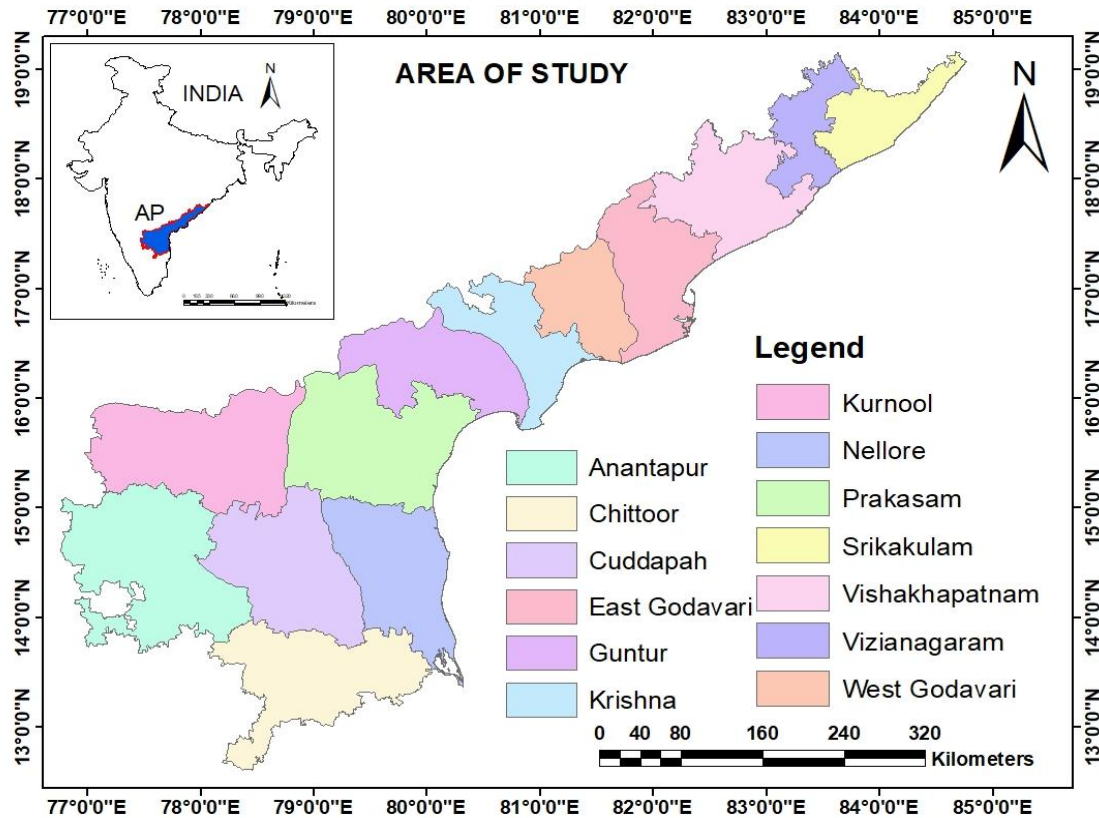


Fig. 1. Map showing the area of study

3.7 Spatial Clustering and Hotspot Identification

Localized spatial autocorrelation was examined through the Getis-Ord G_i^* statistic. This analysis employed a 3×3 pixel moving window to identify statistically significant clusters of elevated LST. High positive G_i^* values were utilized to delineate thermal hotspots across the study area.

3.8 Class-Specific Thermal Analysis

Mean LST values were extracted for individual ESA WorldCover classes. To enhance interpretability, these were consolidated into seven primary categories such as Trees, Shrubland, Grassland, Cropland, Built-up, Bare land, and Water bodies.

3.9 Statistical Evaluation and Trend Detection

To determine the relationship between LST and its drivers (NDVI, NDWI, NDBI, and topography), Pearson correlation coefficients were calculated using 4,000 random samples from the 2022 dataset. Furthermore, the non-parametric Mann-

Kendall test was utilized to assess the 2020-2024 time series for monotonic trends, deriving the S statistic, Z-score, and p-values.

This multifaceted framework integrates established remote sensing protocols with advanced spatial statistics to provide a robust assessment of regional thermal dynamics.

4. Results

4.1 Temporal Fluctuations in LST

The analysis of annual mean Land Surface Temperature (LST) across the study region revealed values oscillating between a minimum of 32.47°C in 2024 and a maximum of 33.89°C in 2020. A significant thermal anomaly was recorded in 2022, during which the mean LST reached 32.55°C, marking a substantial reduction of 0.82°C relative to 2021 (Table 2, Fig. 2).

This 2022 decline served as a temporary interruption to a generally stable thermal baseline. Subsequent years showed a partial thermal rebound in 2023 to 33.21°C, followed by a secondary decline in 2024. Over the five-year

observation window, an aggregate variation of approximately 1.42°C was identified, suggesting that transient environmental or anthropogenic fluctuations are superimposed upon an underlying, subtle warming trajectory.

4.2 Correlation Analysis of Biophysical and Topographic Drivers

Examination of the scatter plots for the year 2022 established distinct associations between LST and the various environmental parameters under investigation (Figs. 3-8). The most significant positive relationship was the Normalized Difference Built-up Index (NDBI), which yielded a correlation coefficient of $r = +0.794$. This strong relationship confirms that intensification in built-up density is a primary factor in the elevation of regional surface temperatures (Fig. 5).

Conversely, biophysical features demonstrated a clear mitigating impact on thermal levels. A pronounced inverse relationship was identified with the Normalized Difference Vegetation Index (NDVI, $r = -0.654$), illustrating the robust cooling

effect provided by biomass (Fig. 3). Similarly, moisture availability served as a thermal regulator, as indicated by the moderate negative correlation with the Normalized Difference Water Index (NDWI, $r = -0.541$; Fig. 4).

Regarding topographic variables, the influences were found to be more subtle yet statistically significant. Elevation demonstrated a negative association ($r = -0.201$), aligned with the vertical lapse rate cooling effects typically observed across the Eastern Ghats (Fig. 7). Notably, the slope of the terrain exhibited a stronger negative correlation ($r = -0.504$). This suggests that steeper gradients are associated with lower LST values, potentially due to superior surface drainage and limited heat storage capacity compared to flatter, low-lying areas (Fig. 8).

As summarised in Table 3, these findings validate the hypothesis that urbanization remains the preeminent driver of thermal variation within the state, followed in significance by vegetation cover and topographic characteristics.

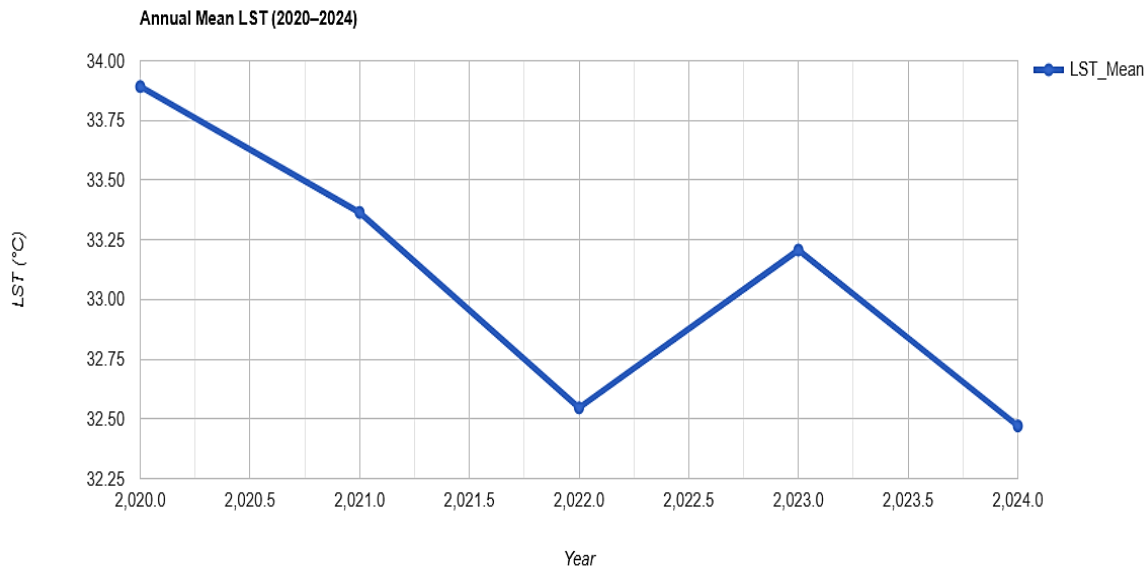


Fig. 2. Spatiotemporal dynamics of annual mean daytime LST in Andhra Pradesh (2020-2024), derived from MODIS MOD11A2 daytime LST composites

Table 2. Annual mean values of LST, NDVI, NDBI, and NDWI across Andhra Pradesh, India (2020-2024), derived from MODIS data

Year	LST Mean (°C)	NDVI mean	NDBI mean	NDWI mean
2020	33.89	0.498	-0.075	-0.075
2021	33.36	0.504	-0.085	-0.085
2022	32.55	0.493	-0.094	-0.094
2023	33.21	0.495	-0.078	-0.078
2024	32.47	0.492	-0.093	-0.093

Table 3. Pearson correlation coefficients (r) between LST and driving factors (NDVI, NDWI, NDBI, elevation, and slope) for 2022 in Andhra Pradesh, India, based on 4,000 random samples from MODIS and SRTM data

Factor	Pearson r with LST
NDVI	-0.654
NDWI	-0.541
NDBI	+0.794
Elevation	-0.201
Slope	-0.504

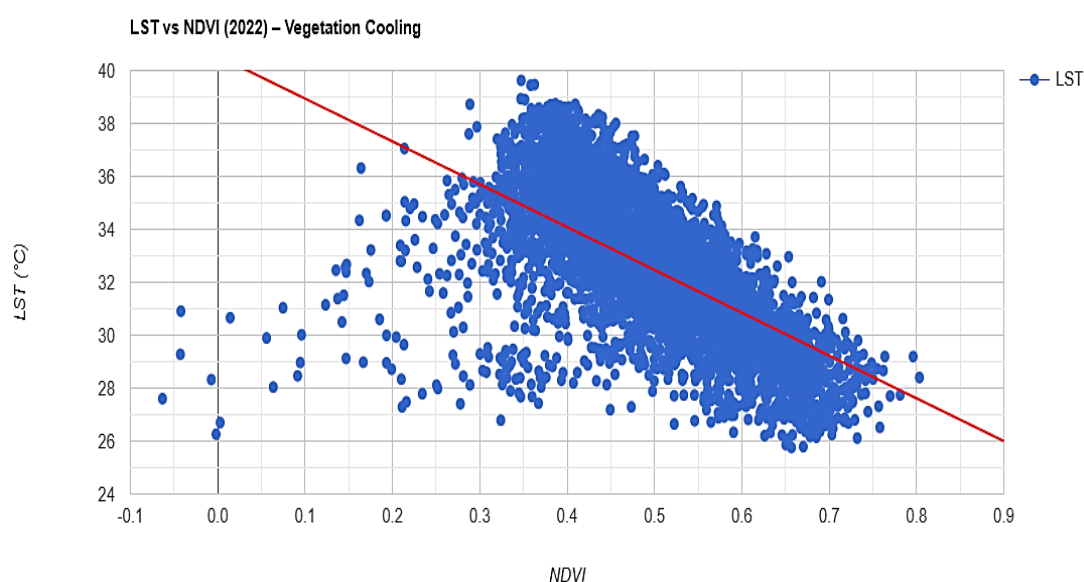


Fig. 3. Relationship between LST and NDVI in Andhra Pradesh for 2022 (n = 5,000 random samples; MODIS data), showing vegetation cooling effect with trend line and R² value

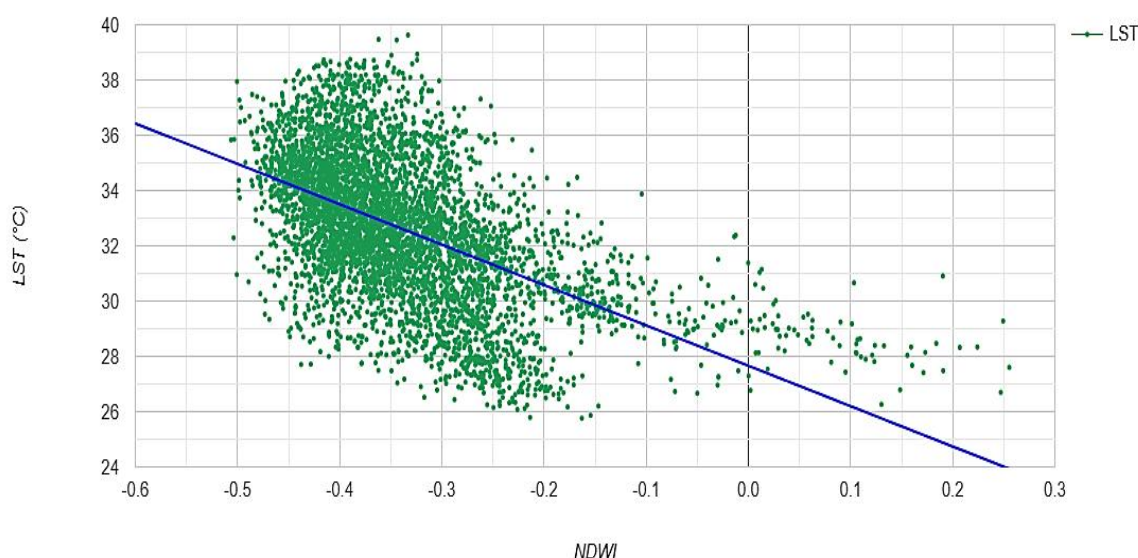


Fig. 4. Relationship between LST and NDWI in Andhra Pradesh for 2022 (n = 5,000 random samples; MODIS MOD09A1 data), illustrating water body cooling effect with trend line and R² value

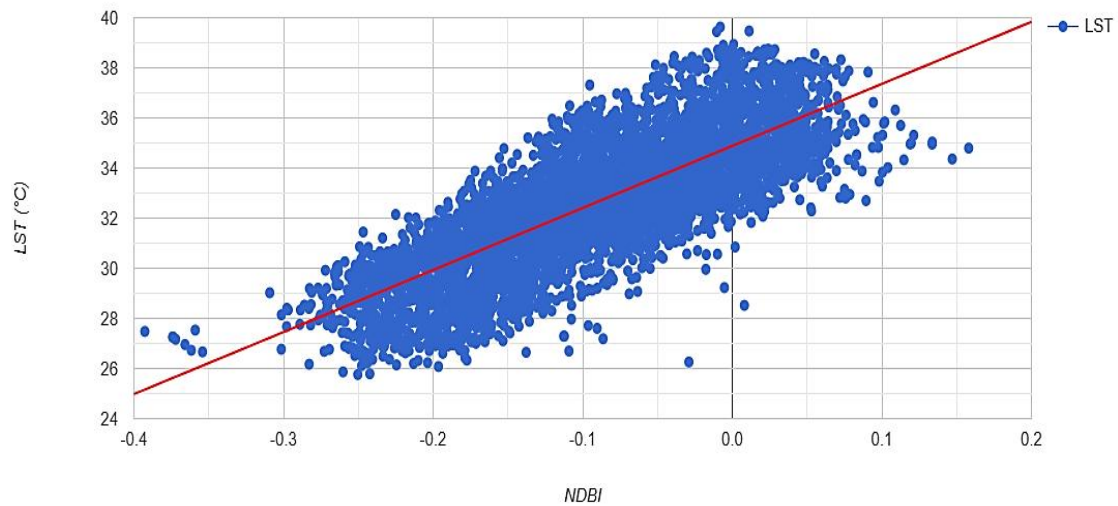


Fig. 5. Relationship between LST and NDBI in Andhra Pradesh, India, for 2022 (n = 5,000 random samples; MODIS MOD09A1 data), demonstrating urban heating effect with trend line and R^2 value

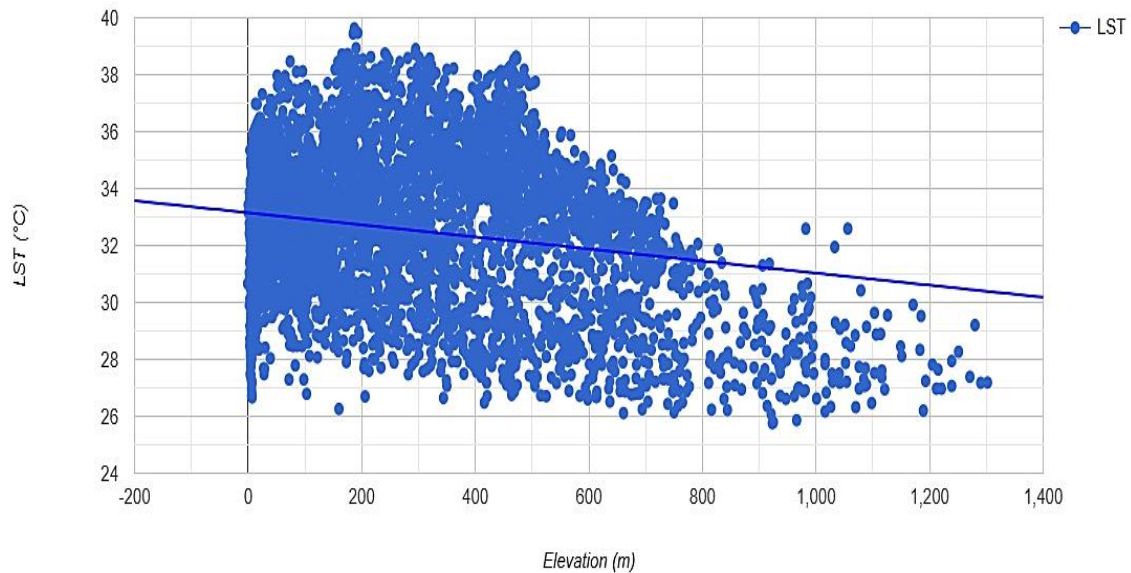


Fig. 6. Relationship between LST and elevation in Andhra Pradesh, India, for 2022 (n = 5,000 random samples; SRTMGL1 DEM and MODIS data), reflecting temperature lapse rate effect with trend line and R^2 value

4.3 Evaluation of Urban Heat Island (UHI) Intensity

The analysis of mean UHI intensity revealed that values remained relatively stable and modest throughout the observation period, fluctuating between a minimum of 0.006 °C in 2024 and a maximum of 0.010 °C during the 2021-2022 interval (Table 4).

This marginal peak observed between 2021 and 2022 was succeeded by a progressive decline toward the end of the study period. This downward trend potentially signifies a period of localized vegetation resurgence alongside the residual effects of curtailed anthropogenic activities during the final phases of pandemic-related restrictions. Despite the low absolute values, the persistence of the UHI effect across all years emphasizes a consistent thermal

disparity between the urban centers and their rural counterparts.

Table 4. Annual mean urban heat island (UHI) intensity in Andhra Pradesh, India (2020-2024), calculated as the difference between actual LST and mean non-urban LST (excluding built-up areas from ESA WorldCover 2021)

Year	Mean UHI intensity (°C)
2020	0.008
2021	0.010
2022	0.010
2023	0.008
2024	0.006

4.4 Spatial Autocorrelation and Hotspot Assessment

Application of the Getis-Ord G_i^* statistics revealed a persistent spatial clustering of elevated LST across the study area. Mean values within these statistically significant hotspots fluctuated between 32.97 in 2020 and a peak of 37.21 in 2022 (Table 5, Figs. 10,11 and Fig. 12).

Notably, the most pronounced thermal clustering was observed in 2022, a year characterised by a general state-level LST anomaly. This indicates that while regional temperatures may have experienced a temporary decline, localized thermal intensification remained robust within identified hotspots. These results suggest that specific urban and industrial corridors maintain

elevated thermal signatures even during periods of broader regional cooling.

Table 5. Annual mean Getis-Ord G_i^* statistic values for land surface temperature clustering in Andhra Pradesh, India (2020-2024)

Year	Mean G_i^* statistic
2020	32.97
2021	34.08
2022	37.21
2023	34.34
2024	36.59

4.5 Evaluation of Land Cover-Specific Thermal Signatures

The analysis of thermal variations across different land-cover categories confirmed that built-up surfaces consistently maintained the highest mean LST throughout the study period. These impervious zones typically exhibited temperatures 2-4 °C higher than those recorded for vegetated classes (Fig. 9).

In contrast, water bodies were identified as the coolest land-cover category, followed by areas characterized by tree cover and shrubland. Cropland and grassland surfaces demonstrated intermediate thermal values. Meanwhile, bare land exhibited notable temperature fluctuations, which were likely associated with seasonal changes in soil moisture and surface exposure. These results further validate the significant impact of surface material composition on regional thermal patterns.

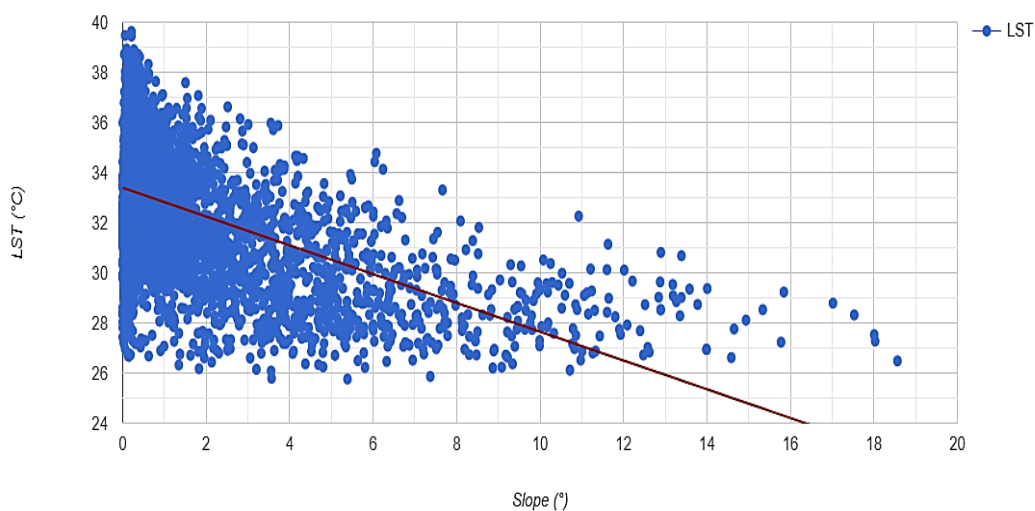


Fig. 7. Relationship between LST and terrain slope in Andhra Pradesh, India, for 2022 (n = 5,000 random samples; derived from SRTMGL1 DEM and MODIS data), indicating topographic influence with trend line and R^2 value

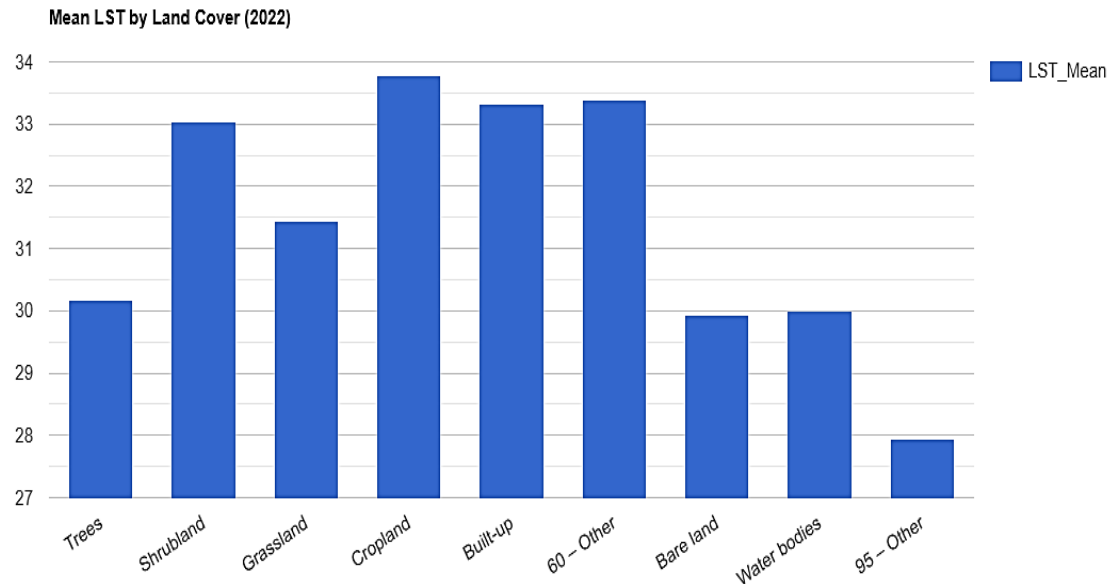


Fig. 8. Mean land surface temperature by land cover class in Andhra Pradesh, India, for 2022 (ESA WorldCover 2021 classification; MODIS LST data), highlighting highest temperatures in built-up areas

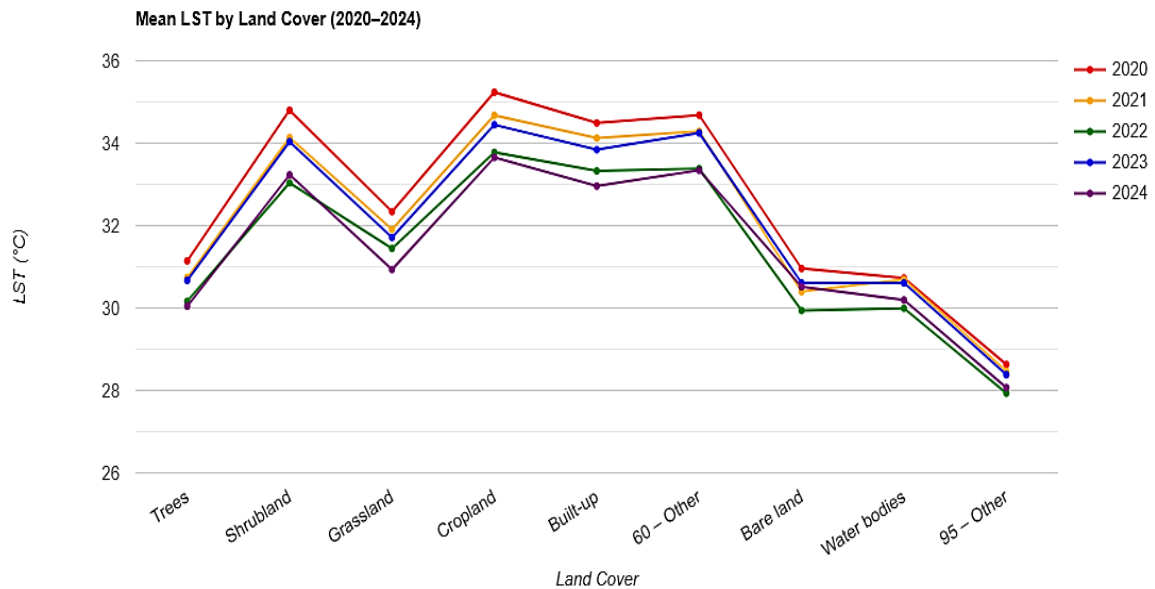


Fig. 9. Inter-annual variation in mean land surface temperature across major land cover classes in Andhra Pradesh, India (2020-2024), based on ESA WorldCover 2021 and MODIS LST data

4.6 Non-Parametric Trend Detection

The application of the Mann-Kendall test to the annual mean LST time series resulted in an S statistic of +10, which represents the maximum possible value for a five-observation sequence. The analysis further produced a Z-score of 2.205 and a p-value of 0.051 (Table 6).

These metrics denote a powerful monotonic increasing trajectory that closely approaches statistical significance at the 95% confidence interval.

In view of the perfect positive pairing observed within the data and the constraints of the short-term time series, the results provide evidence of a robust warming signal throughout the 2020-

2024 period. This trend suggests that despite inter-annual fluctuations, the regional thermal

environment is experiencing a consistent upward shift.

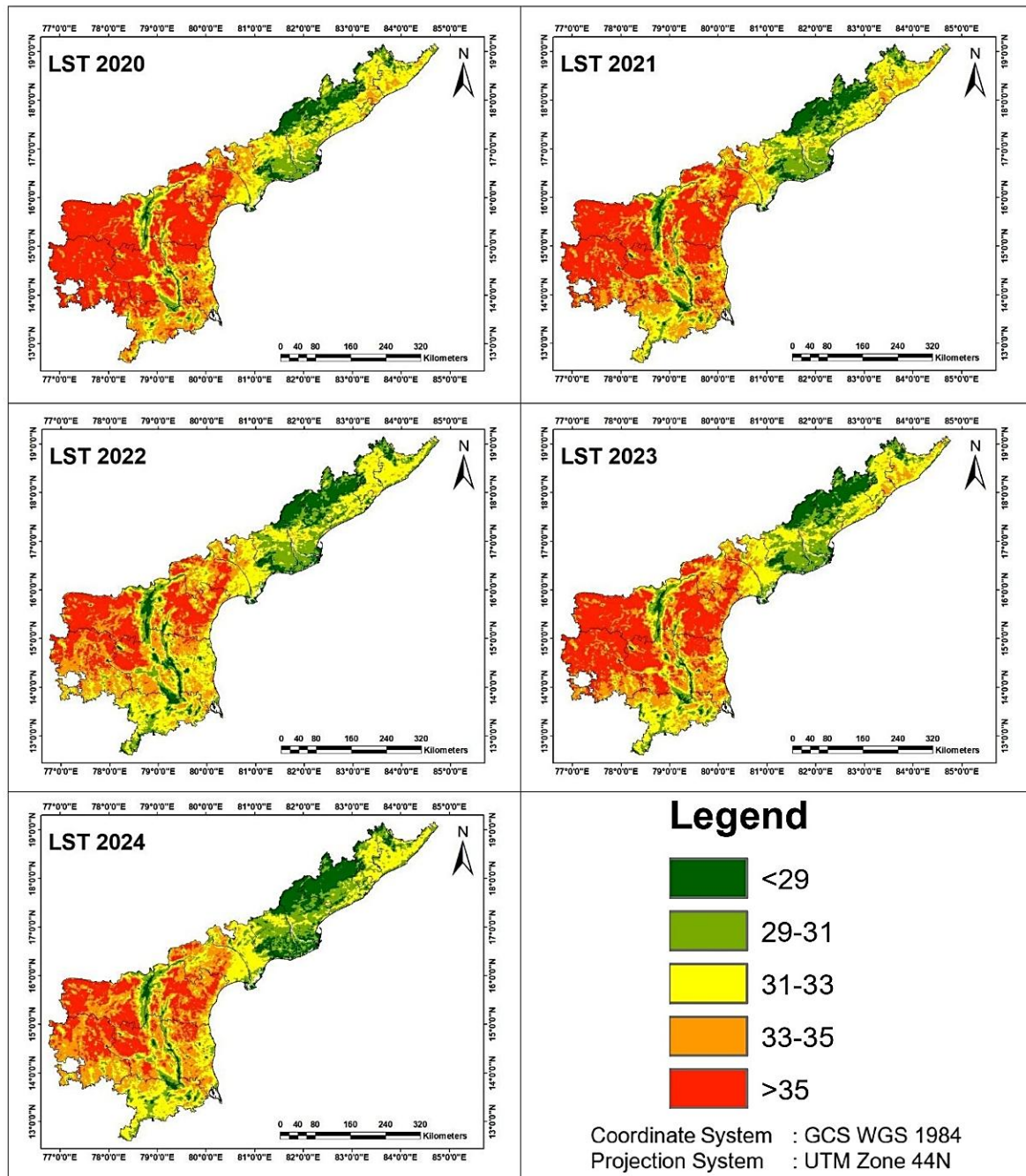


Fig. 10. Spatial distribution of LST for selected years in Andhra Pradesh, India (2020-2024)

The results demonstrate that, notwithstanding a transient cooling anomaly observed in 2022, Andhra Pradesh was subject to persistent thermal pressure throughout the study period. This phenomenon was primarily

catalyzed by the intensification of urban expansion, although its magnitude was modulated by the varying influences of vegetative cover, surface water bodies, and regional topographic characteristics.

Table 6. Mann-Kendall trend test result for annual mean land surface temperature across Andhra Pradesh, India (2020-2024)

Parameter	Value
Period	2020-2024
S Statistic	+10
Z Score	2.205
p-value (approximate)	0.0508
Significance (5%)	No
Trend Direction	Increasing

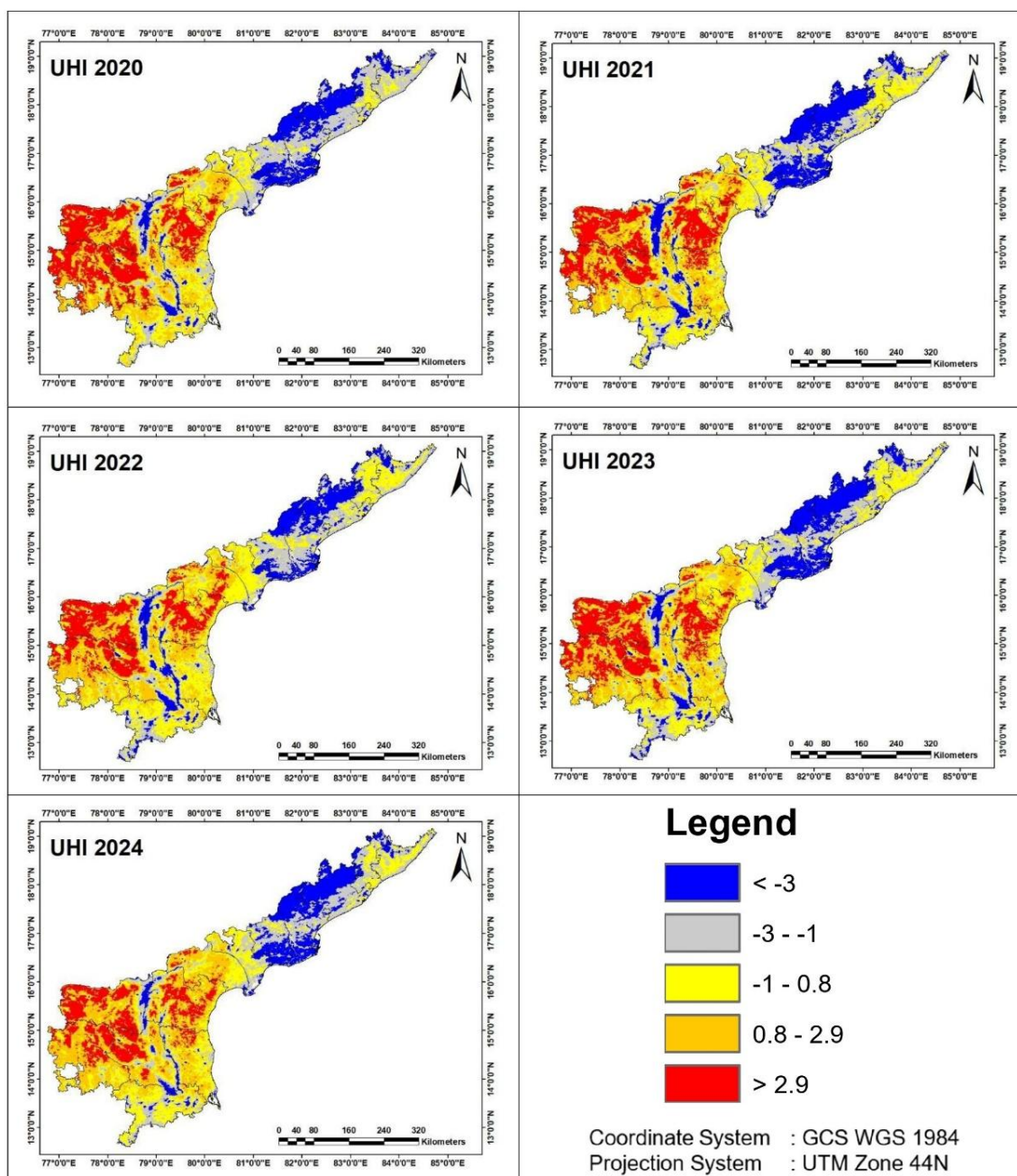


Fig. 11. Spatial distribution of urban heat island (UHI) intensity for selected years in Andhra Pradesh, India (2020-2024)

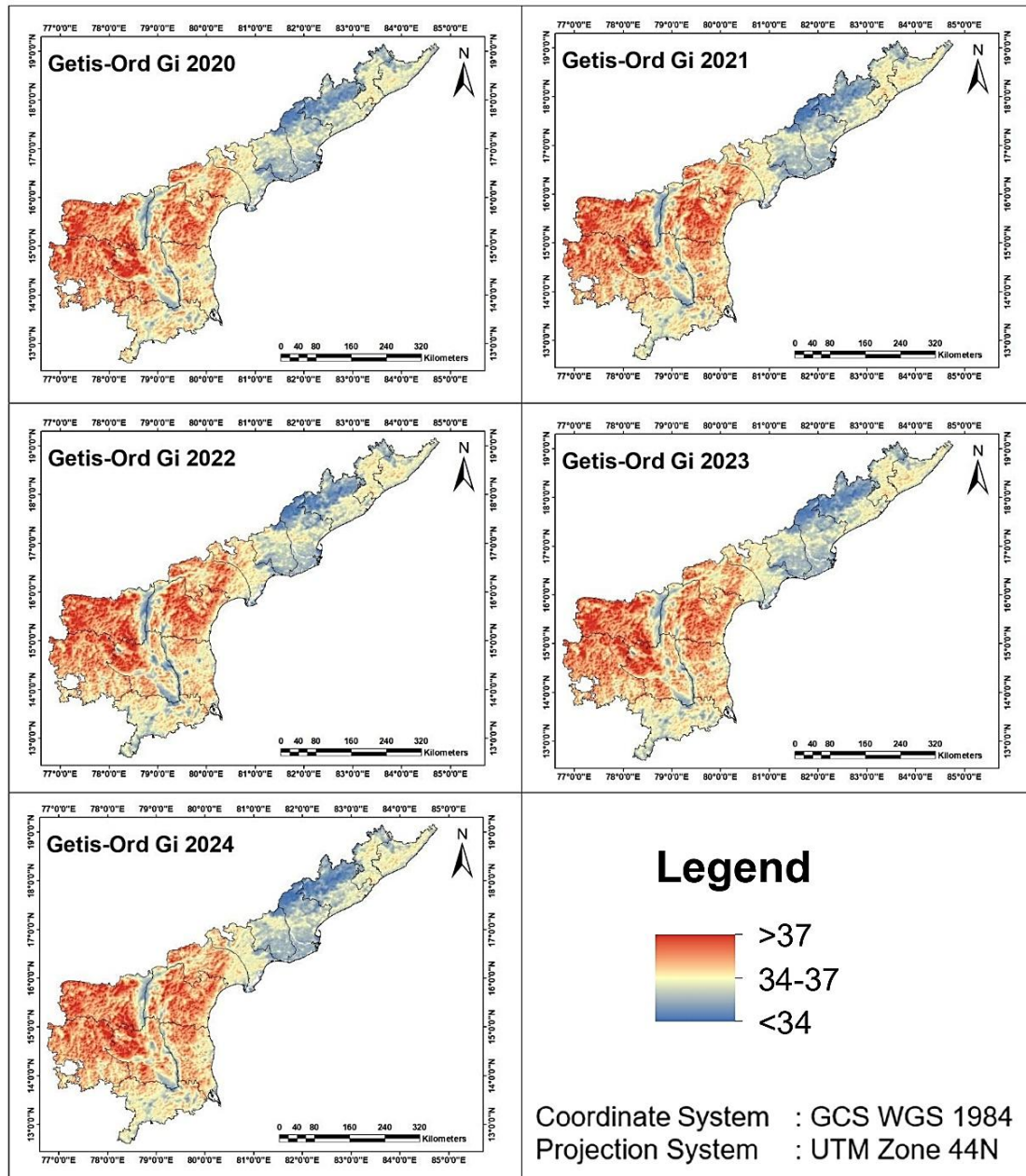


Fig. 12. Spatial distribution of Getis-Ord Gi* hotspots for selected years in Andhra Pradesh, India (2020-2024)

It was further evident that the interplay between these biophysical and anthropogenic factors dictates the state's thermal profile. While urbanization remains the predominant forcing agent, the cooling contributions of natural land covers and elevation-related lapse rates serve as critical regulatory mechanisms. Consequently, the findings suggest that the overarching warming trajectory persists even when short-term environmental fluctuations temporarily suppress regional mean temperatures.

5. Discussion

5.1 Temporal Variability and the 2022 Thermal Anomaly

The findings offer a detailed perspective on LST fluctuations in Andhra Pradesh from 2020 to 2024, identifying a dual pattern of transient oscillations and an underlying upward monotonic trajectory. The distinct thermal decline recorded in 2022 (-0.82°C relative to the 2021) is

consistent with the enhanced vegetative cover and the suppression of anthropogenic activity during the concluding phases of COVID-19 restrictions. Furthermore, optimal monsoon patterns during this interval likely augmented soil moisture levels and latent heat flux through evapotranspiration. Parallel observations of short-term cooling have been documented across various Indian territories during the pandemic (Lele et al., 2021),(Maithani et al., 2020), reinforcing the premise that diminished industrial and vehicular emissions, when coupled with ecological recovery, can temporarily mitigate regional thermal stress.

5.2 Urbanization as the Preeminent Thermal Driver

The resurgence of mean LST during the 2023-2024 period, substantiated by the Mann-Kendall results ($S = +10$, $Z = 2.205$, $p = 0.051$), suggests a transition back to the established warming trend necessitated by urban sprawl. While the p -value slightly exceeds the conventional 0.05 threshold, the attainment of the maximum possible S statistic for a five-year sequence provides compelling evidence of a consistent warming signal, especially when aligned with rising NDBI metrics. Within environmental monitoring, short-duration time series often face statistical power constraints, leading researchers to interpret $p < 0.10$ as indicative of substantive shifts (Nayak et al., 2023). This observation aligns with broader national assessments where urban-induced warming persists as a dominant feature despite intermittent cooling events (Nayak et al., 2023).

The overwhelming correlation between NDBI and LST ($r = +0.794$) highlights the decisive impact of urbanization. Built-up surfaces not only yielded the highest thermal means but also exhibited the most rigorous statistical association with LST, confirming that the replacement of natural substrates with impervious materials is the fundamental mechanism driving regional warming. This outcome corroborates previous site-specific research in Visakhapatnam (Nikkala et al., 2022) and the state's semi-arid regions (Pradeep Kumar et al., 2023), where NDBI served as robust indicators of thermal stress.

5.3 Modulating Effects of Biophysical and Topographic Factors

Conversely, vegetation (NDVI, $r = -0.654$) and aquatic bodies (NDWI, $r = -0.541$) acted as

significant thermal buffers, likely due to their capacity for evapotranspiration and latent heat flux. Topographic influences, while secondary, provided essential modulation. The inverse relationship with slope ($r = -0.504$) suggests that steeper terrain configurations may hinder heat retention through improved drainage, while the more attenuated elevation correlation ($r = -0.201$) is likely a reflection of the state's limited altitudinal gradient across its coastal plains.

5.4 Spatial Clustering and Planning Implications

Although UHI intensities remained relatively modest (0.006 - 0.010 °C), reflecting a more decentralized urban footprint compared to India's megacities, the Getis-Ord G_i^* analysis identified persistent spatial clusters of elevated LST. These hotspots were remarkably distinct in 2022, indicating that localized thermal intensification can persist even during years of regional cooling. Land-cover assessments further established a clear thermal hierarchy, with artificial surfaces consistently maintaining a 2 - 4 °C temperature over natural classes.

These dynamics suggest that urbanization remains the dominant factor shaping the thermal landscape of Andhra Pradesh, outweighing the cooling contributions of topography and moisture. This necessitates the integration of heat-resilient strategies, such as the expansion of "blue-green" infrastructure, in rapidly developing hubs like Visakhapatnam and Vijayawada.

5.5 Limitations and Future Directions

The current study is bounded by the 1 km spatial resolution of MODIS data, which may obscure fine-scale intra-urban thermal variations, and a five-year window that limits the long-term predictive power of trend statistics. Future research should utilize high-resolution datasets, such as Landsat-9 or Sentinel-2, to facilitate sub-urban analysis and extend the temporal scope to refine trend detection. Nevertheless, this study advances the regional LST study in India by synthesizing spectral indices, topographic drivers, and spatial statistics into a reproducible framework for monitoring developing states.

6. Conclusions

The present study has systematically evaluated Land Surface Temperature (LST) configurations across Andhra Pradesh, India, throughout the

2020-2024 interval. By synthesizing biophysical, anthropogenic, and topographic drivers through MODIS data and the Google Earth Engine environment, several critical insights were established. They are,

1. **Temporal variability:** Annual mean LST demonstrated significant inter-annual fluctuations, specifically characterized by a 0.82 °C thermal decline in 2022. This anomaly is largely ascribed to expanded vegetative cover and suppressed anthropogenic forcing during the residual phases of COVID-19 restrictions, prior to a thermal rebound in the subsequent years.
2. **Driving mechanisms:** Pearson Correlation assessments for 2022 pinpointed the Normalized Difference Built-up Index (NDBI) as the dominant positive driver of LST ($r = +0.794$), confirming urban density as the primary regional warming agent. Conversely, the Normalized Difference Vegetation Index (NDVI, $r = -0.654$) and the Normalized Difference Water Index (NDWI, $r = -0.541$) acted as substantial cooling influencers. Topographic factors remained secondary, with slope gradients ($r = -0.504$) and elevation ($r = -0.201$) demonstrating inverse associations with surface temperature.
3. **Spatial and heat island dynamics:** While Urban Heat Island (UHI) intensities were found to be modest (0.006 - 0.010 °C), reflective of the state's decentralized urban footprint, Getis-Ord G_i^* statistics confirmed that statistically significant thermal hotspots persist within major metropolitan corridors.
4. **Land-cover thermal hierarchy:** Comparative analysis of land-use classes consistently identified impervious built-up zones as the warmest surfaces across all observation years, whereas aquatic and forested environments maintained the lowest thermal signatures.
5. **Trend trajectory:** The Mann-Kendall test indicated a robust monotonic increasing trend in annual LST ($S = +10$, $Z = 2.205$, $p = 0.051$). Although the p-value placed at the threshold of traditional significance, the maximum positive S statistic for the five-year series reinforces an underlying warming signal that persists despite short-term environmental fluctuations.

The findings reveal that urban growth serves as the dominant driver of the thermal landscape in

Andhra Pradesh, despite the localized cooling benefits provided by vegetation and topography. These results provide an empirical foundation for urban planning authorities to implement heat-resilient strategies, such as the strategic integration of “blue-green” infrastructure, to mitigate thermal stress in rapidly industrializing coastal regions.

While the spatial resolution of MODIS and the five-year temporal window present certain constraints, this study establishes a robust, reproducible framework for regional thermal monitoring. Future research, utilizing higher-resolution sensors, such as Landsat-9 or Sentinel-2, will be essential for detailing intra-urban thermal configurations and strengthening long-term trend detection.

Data and Code Availability

All the datasets utilized in the present study can be accessed through Google Earth Engine (GEE) which are made available from **NASA LP DAAC** and **USGS** repositories.

The computational framework, such as scripts for Land Surface Temperature (LST) retrieval, spectral index calculation, and spatial analysis, was developed using the GEE JavaScript API. The complete source code is made available in a public repository, (**GitHub Repository:** <https://github.com/suneelcs1615/GoogleEarthEngine/blob/main/LST.js>) to ensure the reproducibility of the findings and to facilitate further regional thermal studies.

Disclaimer (Artificial Intelligence)

Author(s) hereby declare that NO generative AI technologies such as Large Language Models (ChatGPT, COPILOT, etc.) and text-to-image generators have been used during the writing or editing of this manuscript.

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Competing Interests

Authors have declared that no competing interests exist.

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